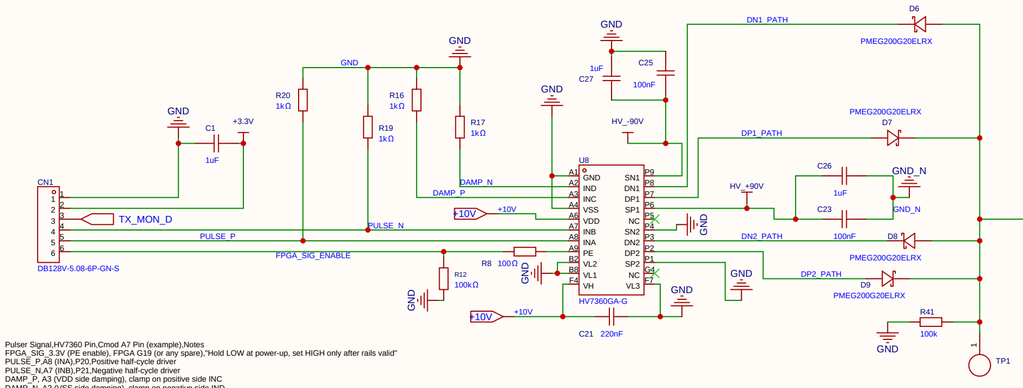
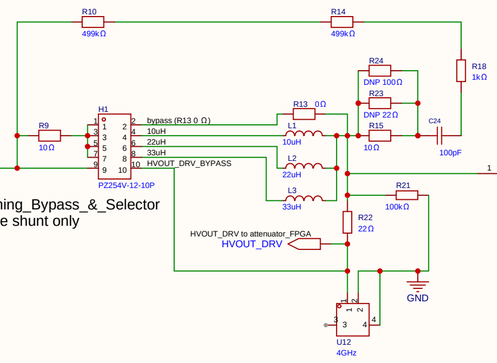
HV Pulser Amp



Pulser Signal,HV7360 Pin,Cmod A7 Pin (example),Notes
FPGA_SIG_3.3V (PE enable), FPGA G19 (or any spare),"Hold LOW at power-up, set HIGH only after rails valid"
PULSE_P,A8 (INA),P20,Positive half-cycle driver
PULSE_N,A7 (INB),P21,Negative half-cycle driver
DAMP_P, A3 (VDD side damping), clamp on positive side INC
DAMP_N, A2 (VSS side damping), clamp on negative side IND
TX_MON_D_(TO_FPGA_INPUT)

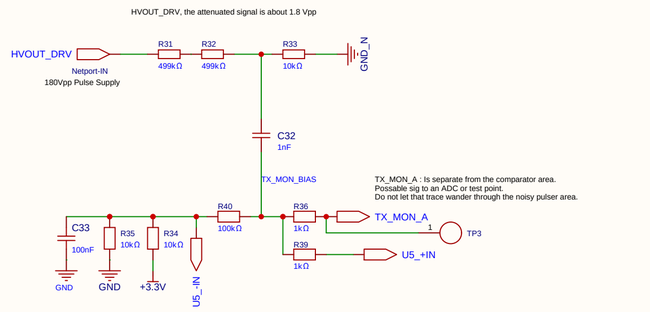
Tuning and impedance matching



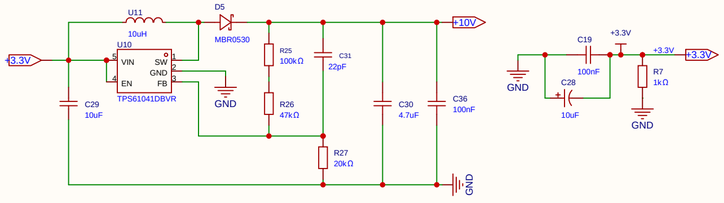
Matching_Bypass_&_Selector
Fit one shunt only

Output. 1.18Mhz 180Vpp to a shear transducer.
BNC connector

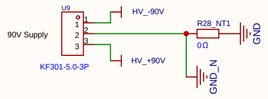
HV pulser sig step down, side (GND_N domain)



3.3 V to ~10.3 V boost supply

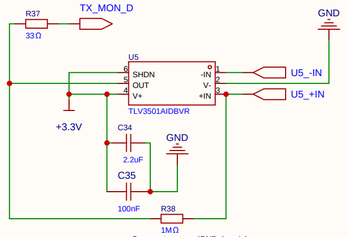


DC-DC Dual 90V Supply connection



Supply from DC-DC 90v supply
U9 Pin 1-90V terminal -> HV_-90V
U9 Pin 2 N terminal -> GND_N / 0V
U9 Pin 3+90V terminal -> HV_+90V

TX_MON_D_(TO_FPGA_INPUT)



Comparator stage (GND domain)

Signal Roles
PULSE_P drives the positive HV pulse toward +90 V
PULSE_N drives the negative HV pulse toward -90 V
DAMP_P / DAMP_N provide a brief 0 V RTZ clamp between pulse halves or at burst end

FPGA to provide:
FPGA_SIG_3.3V, PE remains LOW until all rails are valid, then goes HIGH to enable the pulser

PULSE-A: A ~423 ns window (half the 1.18 MHz period).
-idle time, 10-50ns while idLe (dead time) to prevent shoot-through between DP 1+2 off & DN 1+2 on.
PULSE-B: A ~423 ns window, delayed by the first half-period.

Use these names:

TX_MON_ATT = raw attenuated node before AC coupling
MID_BIAS = 1.65 V bias node
TX_MON_BIAS = biased analog monitor waveform
TX_MON_A = analog monitor output
TX_MON_D = comparator digital output to FPGA

Phase	Signal	State	Duration (ns)
1	PULSE-A	High	413 ns
2	Both	Low	10 ns
3	PULSE-B	High	413 ns
4	Both	Low	11.5 ns
total			847.5 ns

Schematic	HV Pulser 180Vpp	Create at	2026-03-13
		Update at	2026-05-19
Board	HV Pluser 180Vpp	Page	P1
Drawn		HV Pulse lite.tbo	
Reviewed			
		Version	V1.0
		Size	A4
EasyEDA		Page 1 Total 1	EasyEDA.com